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SUPPLEMENTARY MATERIAL
for
**Universal, a Full Periodic Table Force Field for Molecular
Mechanics and Molecular Dynamics Simulations**

A. K. Rappé*, C. J. Casewit, K. S. Colwell, W. A. Goddard III, and W. M. Skiff

**Experimental Bond Distances or Radii (Å), Reference Compound
and Literature Source**

H	1.112; fit so that the C-H distances of propane are 1.112Å.
H__b	1.320; fit to the diborane boron bridging hydride distance. Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. <i>J. Phys. Chem. Ref. Data</i> 1979 , 8, p 630.
He4+4	1.49; extrapolated.
Li	Radius 1.336; <i>Li</i> ₂ ; Huber, K. P.; Herzberg, G. <i>Constants of Diatomic Molecules</i> , Van Nostrand Reinhold; New York, 1979, Vol. 4, p 374.
Be3+2	1.83; <i>Dimethylbis(quinuclidine)beryllium</i> ; Whitt, C.D.; Atwood, J. L. <i>J. Organomet. Chem.</i> 1971 , 32, 17.
B_3	1.590; <i>Tetramethyl diborane</i> ; Hellwege, K-H. <i>Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology</i> ; Springer Verlag: Berlin, 1976; Vol. 7, p 297.
B_2	1.578; <i>Trimethyl borane</i> ; Hellwege, K-H. <i>Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology</i> ; Springer Verlag: Berlin, 1976; Vol. 7, p 254.
C_3	1.526; fit to C_3-C_3 of <i>propane</i> bond order = 1; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. <i>J. Phys. Chem. Ref. Data</i> 1979 , 8, 700.
C_R	1.399; fit to C_R-C_R of <i>benzene</i> bond order = 1 1/2; Hellwege, K-H. <i>Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology</i> ; Springer Verlag: Berlin, 1976; Vol. 7, p 667.
C_2	1.336; fit to C_2-C_2 of <i>propene</i> bond order = 2; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. <i>J. Phys. Chem. Ref. Data</i> 1979 , 8, 697.
C_1	1.207; fit to C_1-C_1 of <i>propyne</i> bond order = 3; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty,

- W. J.; Maki, A. G. *J. Phys. Chem. Ref. Data* **1979**, *8*, 692.
- N_3 1.462; fit to C_3-N_3 of *dimethylamine* bond order = 1; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. *J. Phys. Chem. Ref. Data* **1979**, *8*, 685.
- N_R 1.459; fit to C_3-N_R of *N-methylformamide* bond order = 1; Hellwege, K-H. *Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology*; Springer Verlag: Berlin, 1976; Vol. 7, p 195.
- N_2 1.247; fit to N_2-N_2 of *dimethyldiazene* bond order = 2; Hellwege, K-H. *Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology*; Springer Verlag: Berlin, 1976; Vol. 7, p 202.
- N_1 1.157; fit to C_1-N_1 *acetonitrile* bond order = 3; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. *J. Phys. Chem. Ref. Data* **1979**, *8*, 675.
- O_3 1.410; fit to C_3-O_3 of *methyl ether* bond order = 1; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. *J. Phys. Chem. Ref. Data* **1979**, *8*, 682.
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- O_R 1.428; fit to C_3-O_R of *methyl vinyl ether* bond order = 1; Hellwege, K-H. *Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology*; Springer Verlag: Berlin, 1976; Vol. 7, p 242.
- O_2 1.222; fit to C_2-O_2 of *acetone* bond order = 2; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. *J. Phys. Chem. Ref. Data* **1979**, *8*, 697.
- O_1 1.16; fit to C_1-O_1 of *chromium hexacarbonyl* bond order = 3.
- F 1.383; *Methyl fluoride*; Harmony, M. D.; Laurie, V. W.; Kuczkowski, R. L.; Schwendeman, R. H.; Ramsay, D. A.; Lovas, F. J.; Lafferty, W. J.; Maki, A. G. *J. Phys. Chem. Ref. Data* **1979**, *8*, 658.
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- S_R 1.714; *Thiophene*; Hellwege, K-H. *Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology*; Springer Verlag: Berlin, 1976; Vol. 7, p 270.
- S_2 2.135; $[CH_3C_5H_4MoS_2]_2$; Rakowski DuBois, M. R.; DuBois, D. L.; VanDerveer, M. C.; Haltiwanger, R. C. *Inorg. Chem.*; **1981**, 20, 3064.
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- K Radius 1.952; K_2 ; Huber, K. P.; Herzberg, G. *Constants of Diatomic Molecules*, Van Nostrand Reinhold; New York, 1979, Vol. 4, p 354.
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- Gd6+3 2.45; interpolated
- Tb6+3 2.44; interpolated
- Dy6+3 2.42; interpolated
- Ho6+3 2.41; interpolated
- Er6+3 2.39; interpolated
- Tm6+3 2.38; interpolated

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Pb3	2.203; <i>Tetramethyl lead</i> ; Wong, C.-H.; Schomaker, V. J. <i>Chem. Phys.</i> 1958 , 28, 1007.
Bi3+3	2.267; <i>Trimethyl bismuth</i> ; Hellwege, K.-H. <i>Landolt-boernstein Numerical Data and Functional Relationships in Science and Technology</i> ; Springer Verlag; Berlin, 1976; Vol. 7, p 255.
Po3+4	2.25; extrapolated
At	2.3; extrapolated
Rn4+4	2.04; extrapolated
Fr	Radius 2.44; extrapolated.
Ra6+2	2.97; extrapolated
Ac6+3	2.68; extrapolated
Th6+4	2.44; extrapolated
Pa6+4	2.42; extrapolated
U_6+4	2.41; <i>U(CH₂Ph)₃Me(Me₂PCH₂CH₂PMe₂)</i> ; Edwards, P. G.; Anderson, R. A.; Zalkin, A. <i>Organometallics</i> 1984 , 3, 293.
Np6+4	2.40; extrapolated
Pu6+4	2.38; extrapolated
Am6+4	2.37; extrapolated
Cm6+3	2.50; extrapolated
Bk6+3	2.48; extrapolated
Cf6+3	2.47; extrapolated
Es6+3	2.45; extrapolated
Fm6+3	2.44; extrapolated
Md6+3	2.42; extrapolated
No6+3	2.41; extrapolated
Lw6+3	2.43; extrapolated